

DART

DYNAMIC ADVERSARIAL REAL-TIME THREAT-EVASION

ThreatGNN · TemporalThreatAttention (GRU h=64) · ManeuverHead · 5-Stage Curriculum · 198,872 params · Val AUC 0.9384 · 84% Survival

01 · INPUT

Scene Graph Construction

NODE FEATURES

Jet (type 0) · Missile (type 1)
Flares $\leq 200\text{px}$ of missile (type 2)
5D: $[x, y, vx, vy, type]$

EDGE FEATURES

Fully connected · $n(n-1)$ directed edges
4D: $[\Delta x, \Delta y, \Delta vx, \Delta vy]$

CONTEXT VECTOR

Per-missile analytic features
5D: $[norm_dist, tti, closing_spd, aoa, mtype]$

K sub-graphs · 1 per missile
max 8 nodes · prevents graph pollution

02 · THREATGNN

Message Passing Interaction Network

NODE_ENC

Linear + LayerNorm
 $5 \rightarrow 64$

EDGE_ENC

Linear + LayerNorm
 $4 \rightarrow 32$

METALAYER ROUND

EdgeModel MLP: $160 \rightarrow 128 \rightarrow 64 \rightarrow 32$
NodeModel MLP: $96 \rightarrow 128 \rightarrow 64 \rightarrow 64$
LayerNorm after every linear layer
Edge input: $src \parallel dest \parallel edge = 160D$

x 3

POOL_HEAD

Mean-pool nodes $\rightarrow$ sub-graph embed
 $64 \rightarrow 32$ per missile

output: 32D embedding per missile
168,560 params

03 · TEMPORAL ATTN

TemporalThreat Attention (GRU)

CONCAT EMBED + CTX

$embed(32) \parallel context(5)$
 $\rightarrow 37D$ input per missile

GRUCELL

Shared across K missiles
Hidden state persists across ticks
Resets on missile count change
input=37, hidden=64 · 19,776 params

ATTENTION SCORER

MLP on GRU hidden states
Softmax $\tau=0.5$ (peaked distribution)
 $64 \rightarrow 32 \rightarrow 1$ · weight per missile

WEIGHTED SUM $\rightarrow Z$

$\sum attn_k \cdot h_k \rightarrow$ scene vector z
 $z \in \mathbb{R}^{64}$

temporal memory · no cross-frame backprop

04 · OUTPUTS

Dual-Head Output

DANGER_HEAD

$z \rightarrow MLP \rightarrow Sigmoid$
danger $\in [0, 1]$
 $64 \rightarrow 16 \rightarrow 1 + Sigmoid$ · 1,073 params

MANEUVERHEAD

$[top_hidden \parallel danger] \rightarrow logits$
NORMAL · BARREL_ROLL · JINKING
FALLING_LEAF · COBRA · IMMELMANN
 $65 \rightarrow 32 \rightarrow 16 \rightarrow 6$ · 2,646 params

MPC HEADING SEARCH

12 candidate $\Delta heading$ evaluated
Project 30 frames forward
kinematically
Select min danger candidate
wall penalty 100px · center gravity 0.1x

best_heading · flare_flag · attn_weights · maneuver_idx

DUAL-QUEUE ASYNC THREADING ARCHITECTURE

PYGAME LOOP

60 fps · physics
render · FSM

state_queue

maxsize=1
non-blocking
skip if full

GNN DAEMON

12-coand MPC
+ real fwd pass
4-21ms/tick

decision_queue

maxsize=1
7-tuple output
non-blocking

FSM EXECUTE

heading · flares
maneuver · HUD

Total params: **198,872** ThreatGNN: **168,560** GRUCell: **19,776** ManeuverHead: **2,646** Survival rate: **84%** Median latency: **20.73ms CPU**

Val AUC: **0.9384** Hardware: **Intel i5-8250U · No GPU** Training frames: **330,134** Curriculum stages: **5**

LOSS FUNCTION

$$\mathcal{L} = \text{BCE}(\text{danger}, y) + 0.4 \times \text{CE}(\text{maneuver}, m)$$

BCE $\rightarrow$ danger regression · binary per frame

CE $\rightarrow$ 6-class maneuver classification

$\lambda=0.4 \rightarrow$ secondary task weight

pos_weight $\rightarrow$ S1: 22.93x $\rightarrow$ S5: 0.37x

Optimizer AdamW lr=3x10⁻⁴ wd=10⁻⁴

Scheduler ReduceLROnPlateau p=2

Grad clip max_norm=1.0

Batch size 256